Midterm Report: Image Captioning and Weather Analysis

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1 Task Overview

Our system generates image captions and predicts weather cues *simultaneously*. A shared CLIP backbone feeds two parallel heads: (1) a CLIP-prefix-to-GPT decoder for natural-language captions, and (2) a weather classifier that enriches the caption with environmental context. Aligning both outputs mitigates the mode-collapse failure we observed when captioning operated alone and ensures user-facing responses reflect scene conditions.

The end-to-end training pipeline couples the caption and weather branches:

1. **COCO dataset** → `01_Database_to_Clip.py`
2. → `CLIP_Pro_train_merged.pkl` / `CLIP_Pro_val_merged.pkl`
3. → `02_Train_Plus.py`
4. → `clip_pro_prefix-best.pt` (model weights)
5. → `03_Predict.py` (caption generation)
6. → `04_Evaluation.py` / `05_Score.py` (metrics)

Joint inference produces captions conditioned on real-time weather predictions; users receive answers that combine textual descriptions with climate indicators (e.g., "A person walking with an umbrella. Weather: rainy (95%).").

The weather branch currently attains ∼40% accuracy. Indoor-heavy validation data and ambiguous sky cues limit performance, motivating targeted data augmentation to balance both tasks.

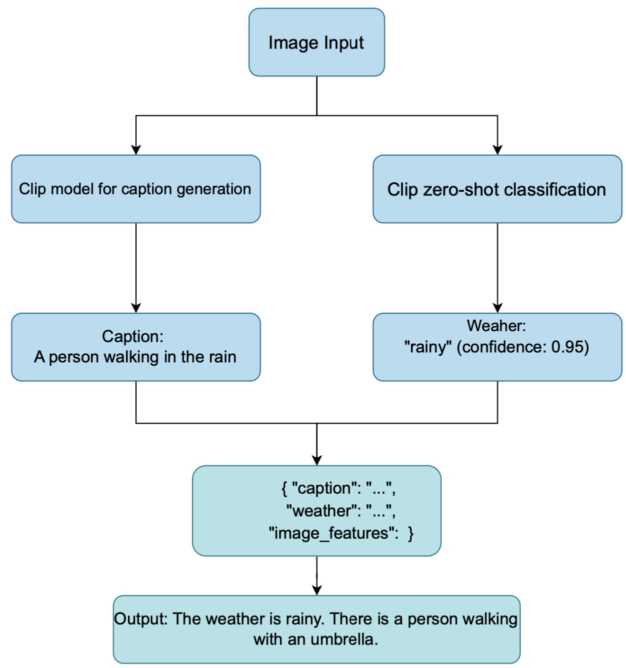

Figure 1: System overview highlighting dual-branch captioning and weather analysis.

2 Background and Related Work

2.1 Contrastive Language-Image Pre-training (CLIP)

CLIP [2] is a vision-language model trained using contrastive loss on 400M image-text pairs collected from the web. It contains a dual-encoder architecture with a visual encoder (ResNet/ViT) and a text transformer. Mokady et al. [2] demonstrate that CLIP embeddings can be projected into the input space of GPT-2, providing a prefix that conditions subsequent caption generation. This motivates our adoption of CLIP as the visual backbone.

2.2 Generative Pre-trained Transformer 2 (GPT-2)

GPT-2 [3] is a transformer-decoder-only architecture optimised for autoregressive language modeling. Its self-attention mechanism captures long-range dependencies more efficiently than recurrent networks. We therefore select GPT-2 as the linguistic decoder in our pipeline, expecting coherent, contextually grounded captions.

2.3 Multi-Head Attention Mechanism

Following Vaswani et al. [4], multi-head attention enables the model to attend to multiple representation subspaces. We employ cross-modal attention between CLIP embeddings and the GPT-2 decoder, allowing the system to focus on salient visual cues when generating each token. This design has proven advantageous in prior captioning frameworks and we observe similar benefits.

3 Experimental Progress

3.1 Caption Generation

We fine-tune the CLIP-prefix-to-GPT-2 mapping using COCO training splits. Preliminary outputs demonstrate syntactically fluent captions, yet content diversity remains limited due to mode collapse. We are exploring:

- Noise injection and augmentation (cf. [1]) to encourage broader caption distributions.
- Temperature and nucleus sampling adjustments during decoding.
- Curriculum learning on curated subsets with higher caption entropy.

3.2 Weather Classification

The weather branch uses CLIP-derived features with a lightweight classifier to predict labels such as *sunny*, *rainy*, or *snowy*. Accuracy plateaued around 40%, likely due to weak weather signals in indoor imagery. Planned improvements include:

1. Augmenting training data with weather-focused datasets.
2. Incorporating temporal cues where available.
3. Applying label smoothing and class re-balancing.

4 CLIP Zero-Shot Weather Classification

Recent advances in zero-shot CLIP classification [8, 9] inspire a complementary evaluation pathway. We implement a CLIP-driven classifier that operates without additional weather labels, generating predictions from textual prompts that describe weather conditions.

4.1 Inference Pipeline

1. **Image encoding**: Extract visual embeddings using the CLIP vision encoder.
2. **Text encoding**: Convert weather descriptions into embeddings with the CLIP text encoder.
3. **Similarity scoring**: Compute cosine similarities between image and text embeddings.
4. **Probability normalization**: Apply softmax to transform similarities into a probability distribution.
5. **Result output**: Return the top-1 weather label alongside class probabilities.

4.2 Prompt Engineering

We enrich the prompt set with descriptive suffixes to emphasise weather cues, following recommendations from prior zero-shot studies. Table 1 lists the prompts currently deployed.

Table 1: Weather categories and associated prompt augmentations used for CLIP zero-shot inference.

Weather type	Prompt suffix
Sunny	*"on a sunny day"*
Rainy	*"in the rain"*
Snowy	*"in the snow"*
Cloudy	*"under cloudy skies"*
Foggy	*"in foggy conditions"*
Stormy	*"during a storm"*
Overcast	*"under overcast skies"*
Clear	*"on a clear day"*

4.3 Output Format

Inference produces: (i) the most probable weather type (e.g., "Rainy"), (ii) confidence scores (e.g., 95%), and (iii) the full probability distribution visualised as a bar chart for interface consumption. These outputs can dynamically adjust caption phrasing to align with environmental context.

5 Next Steps

- Address mode collapse by experimenting with contrastive regularizers and diverse beam search variants.
- Enhance weather detection through specialized datasets and multimodal feature fusion (e.g., integrating horizon detectors or sky segmentation).

References

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